Self-assessment Worksheet (Worksheet 2)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Know the definitions of bounded, convergent and divergent sequences.	☐
• Given that a sequence converges to some value, prove that it converges to that value directly from the definition.	☐
• Be able to create and draw pictures of convergent, divergent and bounded sequences.	☐
• Utilise the definition of a convergent sequence to prove more general results.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Using the definition of a convergent sequence, prove that
  (a) $\lim\left(5-\frac{3}{n}\right)=5$.
  (b) $\lim\left(\frac{n+4}{n+6}\right)=1$.
  (c) $\lim\left(\frac{3\cos(n)}{n+1}\right)=0$.
  (d) $\lim\left(\frac{7\sin(n)}{n}\right)=0$.
  (e) $\lim(a_n)=c\in\mathbb{R}$ where $a_n=c$ for all $n\in\mathbb{N}$.

Question 2 I can do this question ☐

- Give an example or prove the impossibility of each of the following requests:
  (a) A sequence with an infinite number of zeros that does not converge to zero.
  (b) A sequence with an infinite number of zeros that converges to a non-zero limit.
  (c) A divergent sequence with the property that for every $n\in\mathbb{N}$, it is possible to find n consecutive zeros somewhere in the sequence.

Question 3 I can do this question ☐

- Using the definition of a convergent sequence prove that

$$\lim\left(7-\frac{1}{\sqrt{n}}\right)=7.$$

Question 4 I can do this question ☐

- Prove or disprove: If (a_n) converges, then the set $\{a_n : n \in \mathbb{N}\}$ has a maximum.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 5 I can do this question ☐

- Let $\lim(a_n) = a$. Prove directly from the definition of a convergent sequence that
  (a) $\lim(5a_n + 3) = 5a + 3$.
  (b) $\lim(|a_n|) = |a|$.
  (c) $\lim(2a_{n+1} + 3a_n) = 5a$.
  (d) $\lim(a_{n+2} - 2a_{n+1} + a_n) = 0$.

  Hint: For (b), you may find the reverse triangle inequality useful.

Question 6 I can do this question ☐

- Let (a_n) and (b_n) be sequences with
$$|a_n| \leq K|b_n|$$
for some $K \in \mathbb{R}^+$ and all $n \in \mathbb{N}$. Prove that if $\lim(b_n) = 0$ then $\lim(a_n) = 0$.

Question 7 I can do this question ☐

- Let (a_n) be a sequence with $\lim(a_n) = a$. Prove that there exists an $N \in \mathbb{N}$ such that
$$|a_n| \leq 1 + |a|$$
for every $n > N$.

Question 8 I can do this question ☐

- Let (a_n) be a sequence with $\lim(a_n) = a > 0$. Prove that
  (a) There exists an $N \in \mathbb{N}$ such that
  $$a_n > \frac{a}{2}$$
  for all $n > N$.
  (b) $\lim\left(\frac{1}{a_n}\right) = \frac{1}{a}$.

Question 9 I can do this question ☐

- Prove that if (a_n) and (b_n) are sequences that both diverge to ∞, then $(a_n + b_n)$ also diverges to ∞.

Question 10 I can do this question ☐

- Using the definition of a convergent sequence, prove that
  (a) $\lim\left(\frac{(-1)^n}{3n^2}\right) = 0$.

(b) $\lim\left(\dfrac{2n-3}{n^2-2n-5}\right)=0.$

(c) $\lim\left(\dfrac{n^2+5n-2}{(n+3)^2+(-1)^n+2}\right)=1.$

(d) $\lim\left(\dfrac{2+ne^{-n}}{3n^2+\sin(n)+1}\right)=0.$

Question 11 I can do this question ☐

- Suppose that a sequence (a_n) converges to 0.001. Prove that finitely many values of a_n are negative.

Question 12 I can do this question ☐

- Let (a_n) be a sequence with $a_n \in \mathbb{Z}$ for all $n \in \mathbb{N}$.
  (a) Is is possible for $\lim(a_n) = 3.5$.
  (b) If $a_n \neq a_m$ for all $n \neq m$, prove that (a_n) does not converge.
  (c) If (a_n) converges, what can be said about this sequence?

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 13 I can do this question ☐

- Let $[[x]]$ be the greatest integer less than or equal to x. For example, $[[\pi]] = 3$ and $[[3]] = 3$. For each sequence, find $\lim(a_n)$ and verify it with the definition of convergence
  (a) $a_n = [[5/n]]$,
  (b) $a_n = [[(12+4n)/3n]]$.

Question 14 I can do this question ☐

- Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection (σ is called a permutation of $\mathbb{N}$). Show that if (a_n) converges to $a \in \mathbb{R}$, then $(a_{f(n)})$ also converges to a. (This shows that any reordering of a convergent sequence still converges to the same point.)

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!